

Name: _____

Final Practice 01

1. Solve the linear equation.

$$20 = 4(x + 3)$$

2. Let function f be defined by the algebraic expression below:

$$f(x) = \frac{x}{5} + 9$$

Evaluate $f(-15)$.

3. Let function f be defined by the algebraic expression below:

$$f(x) = \sqrt[5]{x - 4} + 3$$

Find the inverse of function f .

4. When a relation is displayed as a curve on an x, y plane:

- What is needed for y to be a function of x ?
- What is needed for x to be a function of y ?
- What is needed for the curve to represent a one-to-one function?

5. Solve the radical equation:

$$7 - \sqrt{x + 10} = 4$$

6. Solve the equation.

$$(8x - 3)(5x + 6) = 0$$

7. State the quadratic formula.

8. Factor the quadratic expression.

$$x^2 - 10x + 9$$

9. Factor the quadratic expression.

$$3x^2 + x - 10$$

10. Expand the quadratic expression. $(x + 6)(x + 7)$

11. Rationalize the denominator. Remember, $i = \sqrt{-1}$.

$$\frac{3 - 2i}{5 - 9i}$$

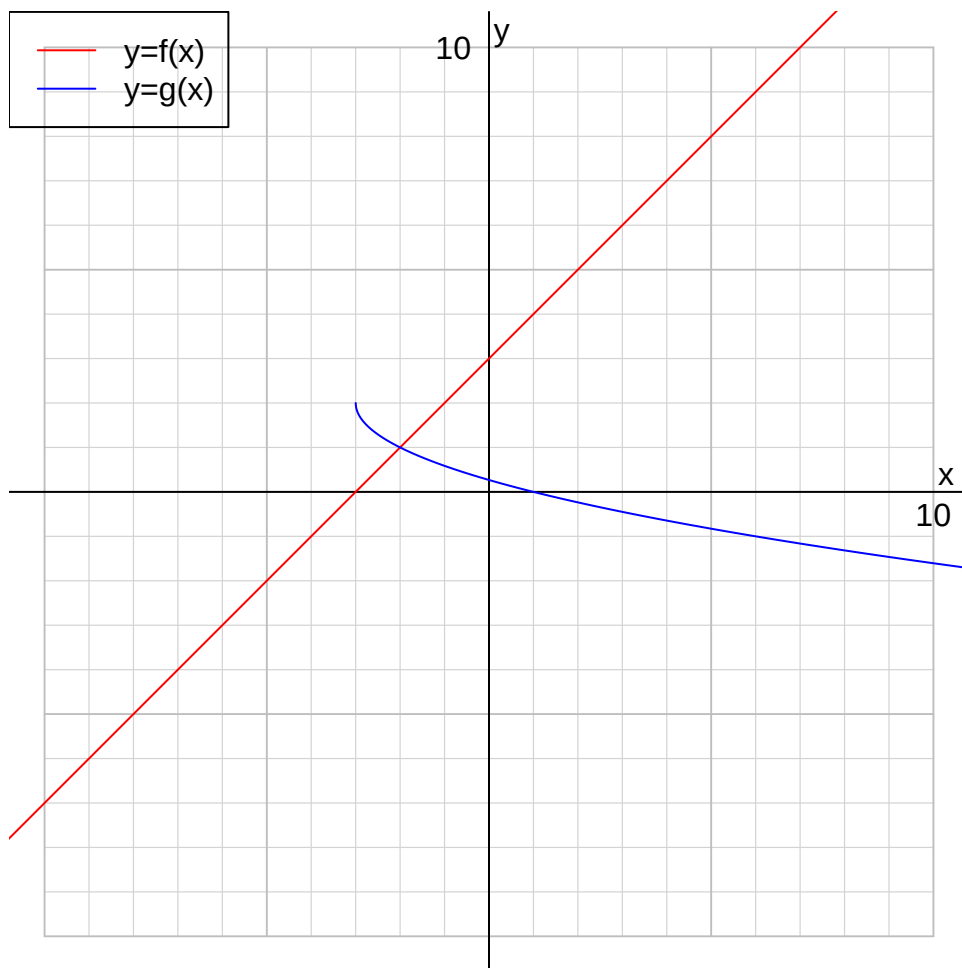
12. Expand the quadratic polynomial with complex roots.

$$(x - (5 + 2i))(x - (5 - 2i))$$

13. Solve the quadratic equation.

$$x^2 - 6x + 13 = 0$$

The questions on this page refer to this graph.



14. Write an equation in slope-intercept form for $f(x)$.

15. Write an equation for $g(x)$.

16. Find the valid solution to $f(x) = g(x)$.

17. Find the extraneous solution to $f(x) = g(x)$.

18. Remember $i = \sqrt{-1}$. Simplify i^{82} .

19. Expand the binomial power expression.

$$(3x + 2)^4$$

20. Expand the product.

$$(x^4 + 2x + 3)(x^3 - 6)$$

21. Use polynomial division to express as a polynomial plus the remainder over the divisor.

$$\frac{2x^3 - 13x^2 + 22x + 2}{x - 3}$$

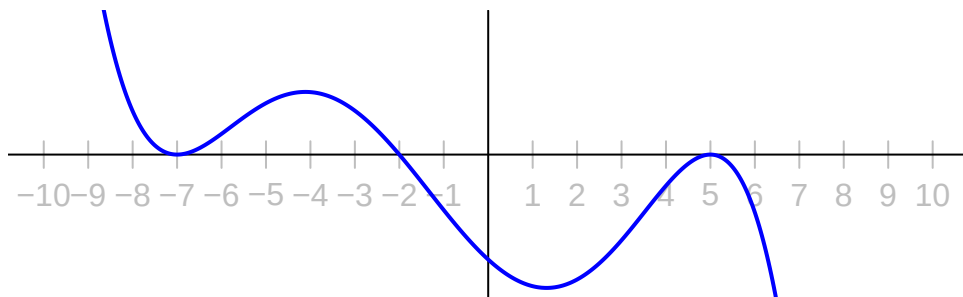
22. If $f(x)$ is a polynomial and $\frac{f(x)}{x-5} = 2x^5 - 3x^3 + 2x + 8$ for all values of x (except $x = 5$), then the remainder theorem promises that $f(5)$ equals what and why?

23. What is the y intercept of the polynomial function $y = 5x^3 - 2x^2 + 3x - 8$?

24. Factor the polynomial $x^3 - 2x^2 - 11x + 12$.

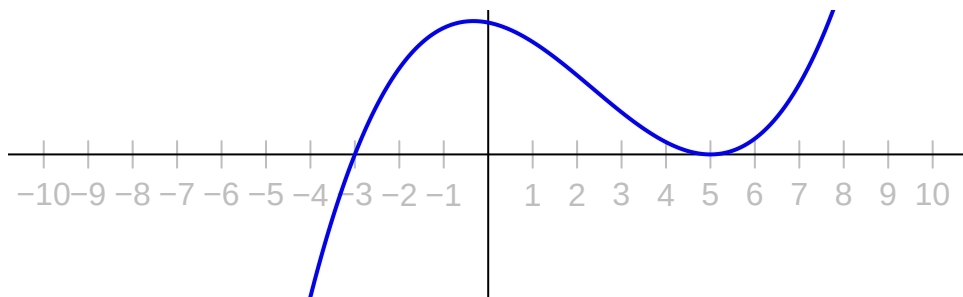
25. Simplify the radical $\sqrt{75}$.

26. A polynomial function is plotted below.



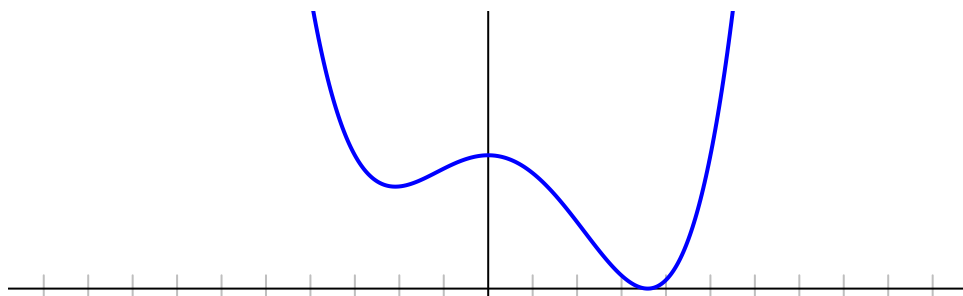
Express the polynomial in **factored** form. Please assume that in standard form the leading coefficient would be 1 or -1.

27. A polynomial function is plotted below.



Express the polynomial in **expanded** form. Please assume that in standard form the leading coefficient would be 1 or -1.

28. A polynomial is plotted below.



What is the minimum degree of the polynomial?

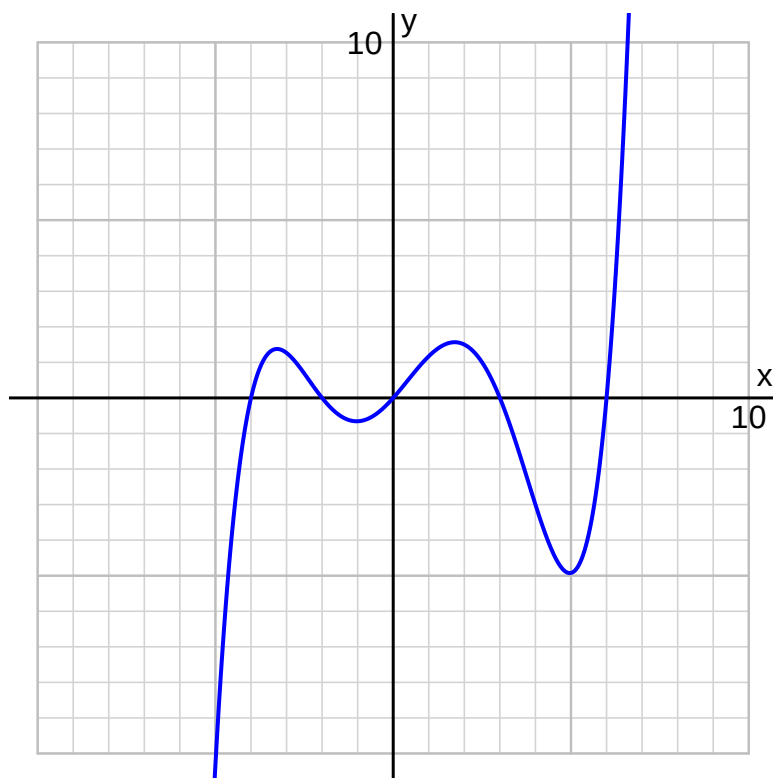
29. The discriminant, $b^2 - 4ac$, is the expression in the radicand in the quadratic formula. Describe the implications of the discriminant being less than 0, more than 0, or equal to 0.

30. Express $(x + 1)^2 - 3(x + 1) - 10$ in factored form.

31. Check if $x - 2$ is a factor of $x^3 - x^2 - 10x - 8$ by using synthetic division.

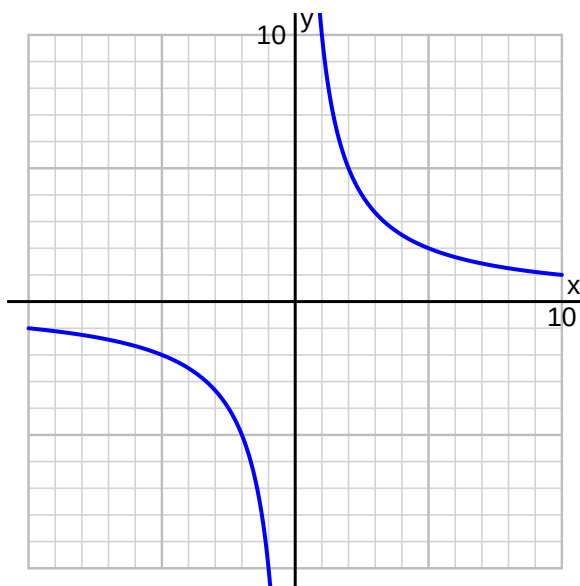
32. Find solution set to $\sqrt{2x + 8} - 3 = x + 1$. Express your answer as a set (listed in curly braces).

33. The graph of $y = p(x)$ below shows a polynomial.



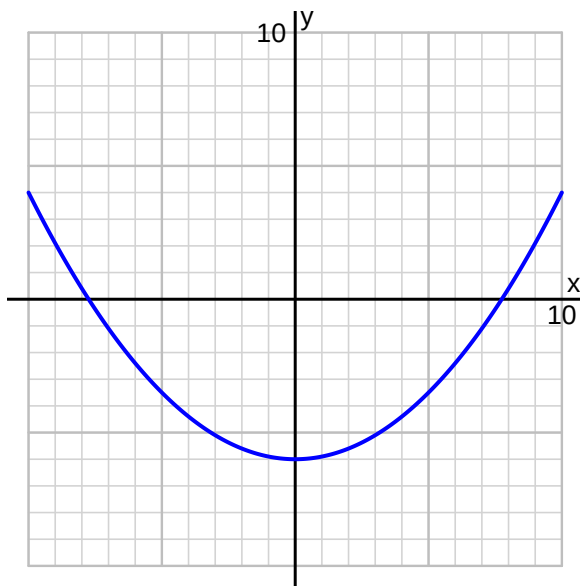
What is the remainder when $p(x)$ is divided by $x - 4$?

34. Consider the function graphed below.



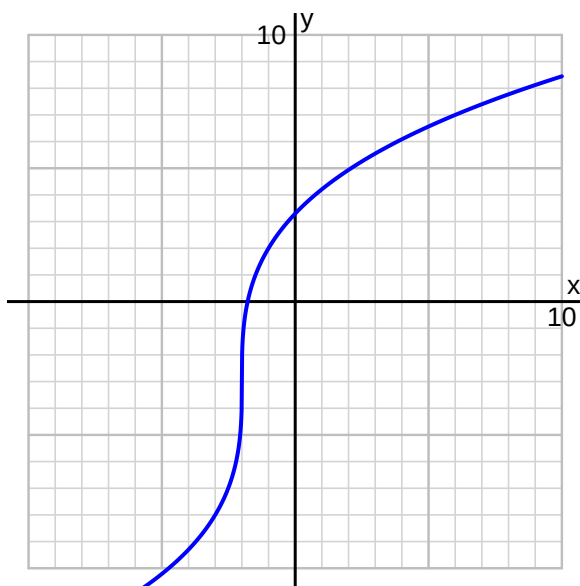
- a. Is the function even?
- b. Is the function odd?
- c. Is the function one-to-one?

35. Consider the function graphed below.



- a. Is the function even?
- b. Is the function odd?
- c. Is the function one-to-one?

36. Consider the function graphed below.

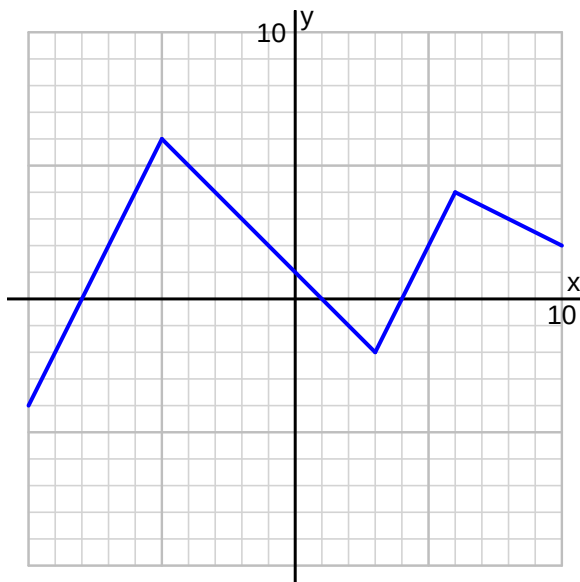


- a. Is the function even?
- b. Is the function odd?
- c. Is the function one-to-one?

37. Find a cubic polynomial with leading coefficient 1 and roots of 2 and $1 + 3i$.

38. Find a cubic polynomial with integer coefficients and a positive leading coefficient that is as small as possible and roots of $\frac{2}{3}$ and $\frac{-3}{4}$ and -2 .

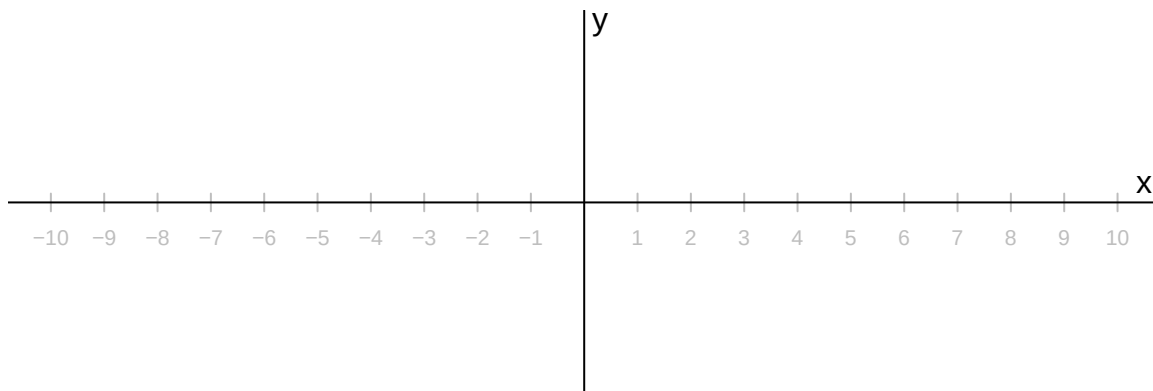
39. Consider the function graphed below. Assume the domain is x values between -10 and 10 (so the entire function is shown here).



- Using interval notation, express where the function is positive.
- Using interval notation, express where the function is negative.
- Using interval notation, express where the function is increasing.
- Using interval notation, express where the function is decreasing.

40. Sketch the polynomial on the axes below. Be accurate with the x intercepts and where the function is positive or negative.

$$y = (x + 3)^2 \cdot (x - 5) \cdot (x + 8) \cdot (x - 1)$$



1. $x = 2$
2. $f(-15) = 6$
3. $f^{-1}(x) = (x - 3)^5 + 4$
4. a. pass vertical line test. b. pass horizontal line test. c. the curve must pass both line tests.
5. $x = -1$
6. $x = \frac{3}{8}$ and $x = \frac{-6}{5}$
7. If $ax^2 + bx + c = 0$ then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
8. $(x - 9)(x - 1)$
9. $(3x - 5)(x + 2)$
10. $x^2 + 13x + 42$
11. $\frac{33+17i}{106}$
12. $x^2 - 10x + 29$
13. $x = 3 \pm 2i$
14. $f(x) = x + 3$
15. $g(x) = -\sqrt{x+3} + 2$
16. $x = -2$
17. $x = 1$
18. -1
19. $81x^4 + 216x^3 + 216x^2 + 96x + 16$
20. $x^7 - 4x^4 + 3x^3 - 12x - 18$
21. $2x^2 - 7x + 1 + \frac{5}{x-3}$
22. Because the remainder when dividing $f(x)$ by $x - 5$ is 0, then we know $f(5) = 0$. This is a special case of the **remainder theorem**.
23. $(0, -8)$
24. $(x - 1)(x + 3)(x - 4)$
25. $5\sqrt{3}$
26. $y = -(x + 7)^2 \cdot (x + 2) \cdot (x - 5)^2$
27. $y = x^3 - 7x^2 - 5x + 75$
28. There are 3 extrema shown, so the minimum degree is 4.
29. If $b^2 - 4ac < 0$, then two complex solutions. If $b^2 - 4ac > 0$, then two real solutions. If $b^2 - 4ac = 0$, then one real solution (with multiplicity 2).
30. $(x + 3)(x - 4)$
31. When you divide $x^3 + 5x^2 + 2x - 8$ by $x - 2$ the result is $x^2 + 7x + 16 + \frac{24}{x-2}$, so the remainder is 24. The remainder is not 0, so $x - 2$ is **not** a factor.
32. The solution set is $\{-4, -2\}$, meaning both $x = -4$ and $x = -2$ are valid solutions to the equation.
33. The **remainder theorem** tells us that $p(4)$ equals the remainder when $p(x)$ is divided by $x - 4$. Look at the graph to find the point $(4, -3)$ on the curve, which tells us $p(4) = -3$, so the remainder is -3.
34. This function is **not** even, but it **is** odd and it **is** one-to-one.
35. This function **is** even, but it **is not** odd and it **is not** one-to-one.
36. This function is **not** even and **not** odd, but it **is** one-to-one.
37. $f(x) = x^3 - 4x^2 + 14x - 20$
38. $f(x) = 12x^3 + 25x^2 - 4x - 12$
39. a. Positive: $(-8, 1) \cup (4, 10)$.
b. Negative: $(-10, -8) \cup (1, 4)$.
c. Increasing: $(-10, -5) \cup (3, 6)$.
d. Decreasing: $(-5, 3) \cup (6, 10)$.
- 40.

